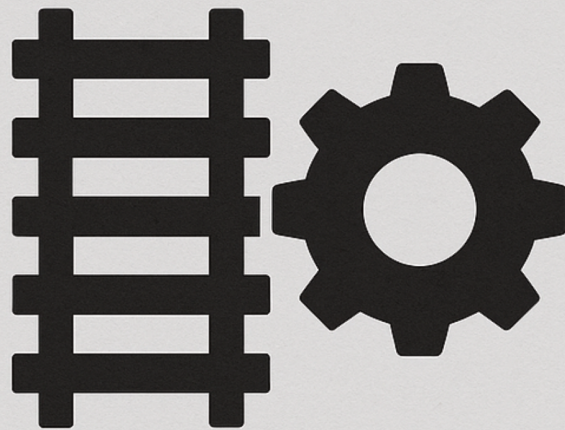


# ACE → DDS Conversion Guide



ACE to PNG to DDS Conversion Tools

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# Chapter 1. Introduction

Legacy Microsoft Train Simulator and Open Rails routes use **.ACE** texture files. These tools provide a simple two-step workflow:

1. Convert **.ACE** files to editable **.PNG** files.
2. Convert the edited **.PNG** files to **.DDS** files for Open Rails.

Most users should use the EXE files from the release **DIST** folder. You do not need Python unless you want to run or change the source scripts.

## Chapter 2. What Is Included

| File                                   | Purpose                                               | Notes                                                                                               |
|----------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <code>ace2png.exe</code>               | Converts <code>.ACE</code> files to <code>.PNG</code> | Self-contained. No Python or ImageMagick is required.                                               |
| <code>png2dds.exe</code>               | Converts <code>.PNG</code> files to <code>.DDS</code> | Does not require Python, but ImageMagick must be installed because it performs the DDS compression. |
| <code>README.md</code>                 | Quick instructions                                    | Included in the <code>DIST</code> folder and in <code>Ace2DDS.zip</code> .                          |
| <code>Documentation\ace2dds.pdf</code> | This user guide                                       | Included with the release package.                                                                  |



The old `PNG2DDS.ps1` PowerShell script is no longer the main tool. PNG to DDS conversion is now handled by `png2dds.exe` / `png2dds.py`.

## Chapter 3. Requirements

### 3.1. If You Are Using the EXE Files

You need:

1. Windows
2. ImageMagick

You do not need Python.

`ace2png.exe` can run by itself. `png2dds.exe` needs ImageMagick to create the final DDS files.

### 3.2. Install ImageMagick

1. Download the Windows installer from:

<https://imagemagick.org/script/download.php#windows>

2. Run the installer.
3. When the installer shows extra options, turn on this option:

Add application directory to your system path

4. Finish the install.
5. Close and reopen your Command Prompt or terminal window.
6. Check that ImageMagick is installed:

```
magick -version
```

If Windows says `magick` is not recognized, reinstall ImageMagick and make sure the PATH option is selected.

## Chapter 4. Basic Workflow

The examples below assume the tools are in `C:\Tools`. You can also place the EXE files directly in the folder where you are working.



If a folder name contains spaces, put quotes around the path. For example:  
`"C:\Open Rails\My Route\Textures"`

### 4.1. Step 1: Convert ACE Files to PNG

Convert one ACE file:

```
C:\Tools\ace2png.exe texture.ace
```

Convert all ACE files in a folder:

```
C:\Tools\ace2png.exe ace_folder -o png_folder
```

Convert ACE files from a route texture folder:

```
C:\Tools\ace2png.exe "C:\ORTSWORLD\My Route\Textures" -o "C:\Work\MyRoute_PNG"
```

### 4.2. Step 2: Edit PNG Files

Open the PNG files in your preferred image editor. Common choices include:

- GIMP
- Paint.NET
- Affinity Photo
- Photoshop

When saving, make sure transparency is preserved if the original texture used transparency. A checkerboard background usually means the image has transparent areas.

### 4.3. Step 3: Convert PNG Files to DDS

Convert one PNG file:

```
C:\Tools\png2dds.exe texture.png
```

This creates a **dds** folder and places **texture.dds** inside it.

Convert all PNG files in one folder:

```
C:\Tools\png2dds.exe png_folder --destination dds_folder
```

Convert all PNG files in a folder and its subfolders:

```
C:\Tools\png2dds.exe png_folder --destination dds_folder --recursive
```

## Chapter 5. What the DDS Converter Does

`png2dds.exe` uses ImageMagick to create DDS files. It chooses the DDS compression automatically:

- Images without transparency use DXT1.
- Images with transparency use DXT5.

When converting folders recursively, subfolders are preserved in the destination folder. This prevents files with the same name in different folders from overwriting each other.

## Chapter 6. Backups

The tools avoid blindly overwriting existing output files.

If an output file already exists, the old file is renamed with `.bak` added to the name. If a backup already exists, the next backup is named `.bak.1`, then `.bak.2`, and so on.

Examples:

```
texture.png  
texture.png.bak  
texture.png.bak.1
```

## Chapter 7. Troubleshooting

| Problem                               | Likely Cause                                                                         | What To Do                                                                                                                              |
|---------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <code>magick</code> is not recognized | ImageMagick is not installed or is not in PATH                                       | Reinstall ImageMagick and select the PATH option. Then open a new terminal window.                                                      |
| No files converted                    | The wrong folder was selected, or the folder does not contain the expected file type | Check that the source folder contains <code>.ACE</code> or <code>.PNG</code> files.                                                     |
| DDS output is missing                 | ImageMagick reported an error or the destination folder cannot be written            | Read the console output and try converting one known-good PNG first.                                                                    |
| Transparency is missing               | The PNG was saved without alpha transparency                                         | Reopen the PNG in your image editor and save it with transparency preserved.                                                            |
| Files are going to the wrong folder   | The destination option was not set as intended                                       | Use <code>--destination dds_folder</code> with <code>png2dds.exe</code> , or <code>-o png_folder</code> with <code>ace2png.exe</code> . |

## Chapter 8. For Developers

To run the Python source scripts directly, install Python 3 and Pillow:

```
pip install Pillow
```

`png2dds.py` also needs ImageMagick installed.

To build both EXE files, regenerate this PDF, stage the `DIST` folder, and create `Ace2DDS.zip`:

```
python build_release.py
```

The release ZIP is created from the `DIST` folder.

## Chapter 9. Credits

ImageMagick is a separate third-party dependency and remains subject to its own license terms.

Referenced Open Rails ACE format documentation © Open Rails Project contributors.

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